PROCTORX: AN AI-POWERED ADAPTIVE ONLINE EXAMINATION SYSTEM WITH EDGE COMPUTER VISION PROCTORING AND 35-DIMENSIONAL LEARNER TELEMETRY

ADAPTIVE AI & EDGE COMPUTING PROJECT

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PROJECT WORK

Submitted by

BAVITHRAN N - 722824106023

BATCH
2024 – 2028

Sri Eshwar College of Engineering
(An Autonomous Institution)
Kinathukadavu (Tk), Coimbatore - 641 202, Tamil Nadu
Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai

BONAFIDE CERTIFICATE

Certified that this Report titled **"PROCTORX: AN AI-POWERED ADAPTIVE ONLINE EXAMINATION SYSTEM WITH EDGE COMPUTER VISION PROCTORING AND 35-DIMENSIONAL LEARNER TELEMETRY"** is the bonafide work of

BAVITHRAN N 722824106023

who carried out the project work under my supervision.

SIGNATURE
Dr. N. Shanmugasundaram, M.E., Ph.D.,
PROFESSOR & HEAD
Department of ECE,
Sri Eshwar College of Engineering,
Coimbatore – 641 202.

SIGNATURE
Dr. S. Ramalingam, M.E., (Ph.D.)
SUBJECT FACULTY
Assistant Professor,
Department of ECE,
Sri Eshwar College of Engineering,
Coimbatore – 641 202.

Submitted for the Autonomous Semester End Project Viva-Voce held on ……...

INTERNAL EXAMINER **EXTERNAL EXAMINER**

QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

INSTITUTE VISION

To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

INSTITUTE MISSION

- Providing outcome and value based engineering education.
- Nurturing research and entrepreneurial culture.
- Enabling students to be industry ready and fulfil their career aspirations.
- Grooming students through behavioral and leadership training programs.

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DEPARTMENT VISION

To groom students into futuristic and globally competent Electronics and Communication engineering professionals.

DEPARTMENT MISSION

- To impart quality education with moral and ethical values to develop competent engineers, leaders and successful entrepreneurs
- To establish state-of-art infrastructure and provide opportunities to update on emerging tools and technologies
- To empower the faculty towards excellence in teaching – learning, consultancy, research and development activities
- To foster socially relevant and industry-oriented innovation among students.

PROGRAM EDUCATIONAL OBJECTIVES

PEO1: Pursue career in multinational organizations, research organizations and core industries, higher studies at premier institutions and establish start-ups.

PEO2: Acquire core competencies in Electronics and Communication Engineering and exposure to latest Electronic Design Automation (EDA) tools.

PEO3: Exhibit professional skills and collaborative work experience.

PROGRAM OUTCOMES

PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)

PO4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

PO8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

PO10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES

PSO1: Interpret and Design Electronic systems using Internet of Things, VLSI Technology and Efficient signal processing algorithms

PSO2: Apply knowledge to solve challenges in Communication Systems and Networks

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ABSTRACT

Online examinations and automated recruitment assessments are widely adopted across educational institutions and technology enterprises. However, traditional evaluation systems suffer from static question delivery and intrusive remote proctoring. Static assessments present identical question sets regardless of a student's competency, enabling question leaks and memorization while failing to diagnose sub-skill deficiencies. Concurrently, conventional remote proctoring mandates high-definition video streaming to central servers, creating severe network bottlenecks, prohibitive bandwidth costs, and serious candidate privacy invasions.

This project proposes **ProctorX**, an AI-powered, multi-modal adaptive online examination and proctoring ecosystem. ProctorX introduces an intelligent Adaptive Testing Engine driven by a comprehensive **35-Dimensional Learner Telemetry Vector** spanning 12 Core Concepts, 9 Algorithmic Patterns, 5 Coding Competencies, and 9 Diagnostic Error Taxonomies. The engine combines **Bayesian Knowledge Tracing (BKT)**, **Item Response Theory (IRT 2PL)**, and a **Multi-Hop Knowledge Graph Decay Ripple Algorithm** to dynamically estimate candidate mastery in real time. Leveraging dedicated **Groq Cloud LPUs running Llama-3.3-70B-Versatile**, the system dynamically synthesizes personalized programming challenges matched precisely to the candidate's cognitive frontier within 1200 ms.

Simultaneously, ProctorX incorporates an ultra-lightweight, **Client-Side Edge Computer Vision Proctoring Pipeline** executed in browser **Web Workers** using **YuNet** and **YOLO ONNX Runtime WebAssembly**. Without transmitting raw video across the network, the client-side system performs real-time face verification, head-pose estimation (yaw, pitch, roll), gaze tracking, multi-face presence detection, and unauthorized object detection (smartphones, books) at 20+ FPS. A sandboxed multi-language compiler pipeline (Piston Engine) executes code in isolated Linux containers for C, C++, Java, and Python. Deployed across React 18 on Vercel, Spring Boot 3 on Render, and Redis Cloud, ProctorX achieves millisecond response times, absolute candidate privacy, and zero server video streaming overhead.

Also this project aligns with multiple United Nations Sustainable Development Goals (SDGs):

- **SDG 4: Quality Education (Target 4.4)** — by equalizing access to technical education through individualized, unbiased, adaptive cognitive assessment.
- **SDG 9: Industry, Innovation, and Infrastructure (Target 9.5)** — through the development of privacy-first edge AI architectures that eliminate bandwidth burdens in developing regions.
- **SDG 8: Decent Work and Economic Growth (Target 8.6)** — by elevating technical recruitment standards through authentic, cheat-resistant coding evaluation.

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CHAPTER 1
INTRODUCTION

Online examinations have become the cornerstone of academic evaluation and global technical hiring. However, conventional digital testing systems remain anchored to rigid, static question delivery. In traditional setups, every candidate receives identical or randomly sampled question sets irrespective of their mastery. This static approach leads to severe evaluation imbalances: proficient students complete non-challenging assessments without testing their upper bounds, while struggling students encounter insurmountable hurdles early on, offering no granular diagnostic insights.

Concurrently, academic integrity in remote environments represents a major challenge. The advent of AI copilots, dual-monitor configurations, and secondary devices has rendered standard lockdown browsers insufficient. Existing commercial proctoring platforms combat cheating through centralized video streaming: webcams, audio streams, and screens are continuously uploaded to remote servers. This brute-force paradigm creates bandwidth bottlenecks, incurs high cloud streaming expenses, and compromises candidate privacy.

The proposed project, **ProctorX**, addresses these challenges by integrating Edge Computing (Client-Side Computer Vision), Dynamic Generative AI (Groq LPUs), and Psychometric Modeling (Bayesian Knowledge Tracing & Item Response Theory) into a unified evaluation platform.

1.1 BACKGROUND AND MOTIVATION

Technical hiring and computer science education require evaluating practical problem-solving rather than rote memorization. Conventional coding assessments present fixed problem sets vulnerable to online leakage and plagiarism. A modern assessment system must adapt in real time to the student's frontier of knowledge.

Proctoring solutions that stream video feeds continuously demand 1.5 to 3.0 Mbps upload bandwidth per candidate. During large-scale university examinations with thousands of concurrent examinees, server bandwidth saturates, leading to disconnections. Furthermore, storing students' home video feeds on cloud servers raises severe data privacy concerns under GDPR and FERPA.

The motivation of ProctorX is twofold: to provide continuous, multi-dimensional cognitive tracking using a 35-dimensional learner vector, and to execute computer vision models entirely on client-side Web Workers, eliminating server video streaming while safeguarding integrity.

1.2 PROBLEM STATEMENT

Existing examination platforms lack the ability to continuously estimate a candidate's latent cognitive ability and synthesize novel, personalized programming problems in real time. Furthermore, existing proctoring mechanisms impose high network bandwidth overheads, introduce latency, and compromise student privacy through centralized video storage.

The problem addressed in this project is to develop an AI-powered, decoupled online examination and proctoring ecosystem that:

- Maintains a continuous 35-dimensional learner telemetry vector modeling concepts, algorithmic patterns, competencies, and error taxonomies.
- Dynamically generates syntactically validated algorithmic challenges via ultra-fast Groq LPU inference (< 1500 ms).
- Executes face verification, gaze estimation, and prohibited object detection locally in browser Web Workers without transmitting video streams over the network.
- Executes untrusted candidate code within secure, sandboxed Linux execution environments.

1.3 CORE OBJECTIVE OF THE PROJECT

Primary objective

To design, implement, and validate ProctorX, an autonomous online examination platform combining edge computer vision proctoring with dynamic LLM question generation and Bayesian Knowledge Tracing.

Specific objectives

- Model candidate competency across 35 telemetry dimensions using Bayesian Knowledge Tracing (P(L0), P(T), P(S), P(G)) and Item Response Theory (θ).
- Implement a Multi-Hop Knowledge Graph Reflection algorithm to propagate mastery updates across adjacent data structure nodes using calibrated decay weights.
- Interface with Groq Cloud LPUs running Llama-3.3-70B to dynamically generate customized coding problems and hidden test suites within 1200 ms.
- Build client-side Web Workers running YuNet (Face/Gaze) and YOLOv8n ONNX (Object Detection) to detect multi-face presence, gaze aversion, and mobile phone usage locally.
- Deploy a decoupled cloud architecture using Spring Boot 3 on Render and React 18 on Vercel with hardened cross-origin security and offline resilience.

1.4 SCOPE AND SIGNIFICANCE

The scope of ProctorX encompasses client-side edge computer vision pipelines, multi-threaded Web Workers, adaptive algorithmic engines, sandboxed code execution, and administrative telemetry consoles. It includes real-time telemetry updates across 12 Data Structures, 9 Algorithmic Patterns, 5 Coding Competencies, and 9 Error Taxonomies.

The system is evaluated under diverse network environments (stable broadband, high-latency, and offline simulations) across multiple programming languages (C, C++, Java, and Python). Edge vision performance is calibrated to maintain $\ge 15\text{ FPS}$ on standard commodity dual-core laptops.

The wider significance of this work includes:

- **Bandwidth Democratization:** Shifting video processing to the edge reduces bandwidth consumption from ~2.5 Mbps to < 10 Kbps, enabling students with rural internet connections to participate in secure remote assessments.
- **Privacy Preservation:** Zero video or audio data leaves candidate memory, eliminating biometric data breach liabilities and ensuring full regulatory compliance.
- **Authentic Evaluation:** Dynamic problem synthesis targeted at the candidate's Zone of Proximal Development eliminates cheating via memorized answer banks.

CHAPTER 2
LITERATURE SURVEY

Digital examination systems combine candidate interfaces, cloud compute services, database management, and integrity verification modules. In recent years, computerized assessments have evolved from simple static form submission portals into complex automated evaluation platforms.

2.1 REVIEW OF EXISTING SYSTEMS AND SOLUTIONS

Traditional Learning Management Systems (Moodle, Blackboard, Canvas)

Traditional LMS platforms deliver pre-authored quizzes and exams. Security relies on client-side browser lockdown tools (e.g., Safe Exam Browser). These systems do not perform biometric verification, gaze tracking, or dynamic question synthesis.

Centralized Remote Proctoring Services (ProctorU, Honorlock, Examity)

Commercial proctoring platforms enforce integrity by streaming candidate webcams, audio feeds, and desktop displays to centralized cloud servers or remote human invigilators. While effective, they introduce high infrastructure costs ($15–$30/hour), severe bandwidth requirements, and substantial student privacy concerns.

Automated Technical Assessment Platforms (HackerRank, LeetCode)

Platforms like HackerRank provide browser-based code editors and sandboxed execution against test cases. However, their questions are static or pulled from fixed pools, making them vulnerable to generative AI copy-pasting, and proctoring is generally limited to browser tab-switch logging.

2.2 LIMITATIONS OF EXISTING SOLUTIONS

Existing online examination solutions exhibit several key limitations:

- **Static Question Sets:** Fixed question banks allow candidates to memorize solutions or leverage second-device AI assistants, undermining assessment authenticity.
- **Severe Bandwidth Demands:** Continuous 720p/1080p video streaming requires 1.5 to 3.0 Mbps per student, causing widespread connection failures on unstable domestic networks.
- **Privacy and Compliance Risks:** Uploading and storing biometric video recordings on corporate cloud servers creates significant GDPR and FERPA compliance liabilities.
- **Binary Scoring:** Conventional platforms evaluate coding solutions solely on Pass/Fail outcomes without analyzing underlying algorithmic reasoning, boundary logic, or complexity tradeoffs.

2.3 COMPARATIVE STUDY ON ALTERNATIVE SOLUTION

Table 2.1 presents a qualitative and functional comparison between existing assessment systems and the proposed ProctorX platform.

Table 2.1: Comparison of online examination and proctoring approaches

Evaluation Metric	Traditional LMS	Commercial Proctoring	Coding Portals	ProctorX Platform
Question Delivery	Static pool	Static pool	Fixed problem bank	**Dynamic Groq LPU synthesis**
Cognitive Modeling	None (raw sum)	None	Elo / Percentile	**35-Dimensional BKT + IRT**
Proctoring Execution	Lock browser only	Server video stream	Tab-switch events	**Client-side Web Workers (YuNet/YOLO)**
Bandwidth Needed	< 5 Kbps	1.5 – 3.0 Mbps	< 20 Kbps	**< 10 Kbps (JSON telemetry)**
Candidate Privacy	High	Very Low (Stored video)	High	**Absolute (Zero video leaves RAM)**
Object Detection	None	Human / Cloud server	None	**Real-time YOLO ONNX on edge**
Code Execution	None	External	Cloud Container	**Sandboxed Piston Engine**

CHAPTER 3
PROPOSED SOLUTION

The proposed ProctorX ecosystem combines three interconnected layers: a client-side React Single Page Application executing edge computer vision pipelines, a Spring Boot 3 modular monolith backend managing mathematical learner models, and distributed cloud services (Groq LPU, Redis Cloud, MySQL, and Piston Sandboxes).

3.1 PROPOSED SYSTEM OVERVIEW

Candidate Client Layer (React 18 SPA)

Provides the candidate examination workspace, featuring Monaco Code Editor, dynamic problem specification panel, and real-time Proctoring HUD. Offloads face detection and object recognition to dedicated Web Workers to ensure a fluid 60 FPS user interface.

Edge Sensory & Vision Processing

Captures webcam and microphone streams locally. YuNet processes facial landmarks and head pose (yaw/pitch) at 20+ FPS. YOLOv8n ONNX scans for unauthorized physical objects (cell phones, books) at 2 FPS. Only lightweight violation events are transmitted to the backend.

Core Backend & Adaptive Engine Layer (Spring Boot 3)

Hosts the Adaptive Engine Service, maintaining the 35-dimensional candidate telemetry vector. Calculates Bayesian Knowledge Tracing updates, Item Response Theory difficulties, and multi-hop graph decay propagation upon every code submission.

3.2 SYSTEM ARCHITECTURE / BLOCK DIAGRAM

Figure 3.1 illustrates the high-level architecture and data flow of the ProctorX platform.

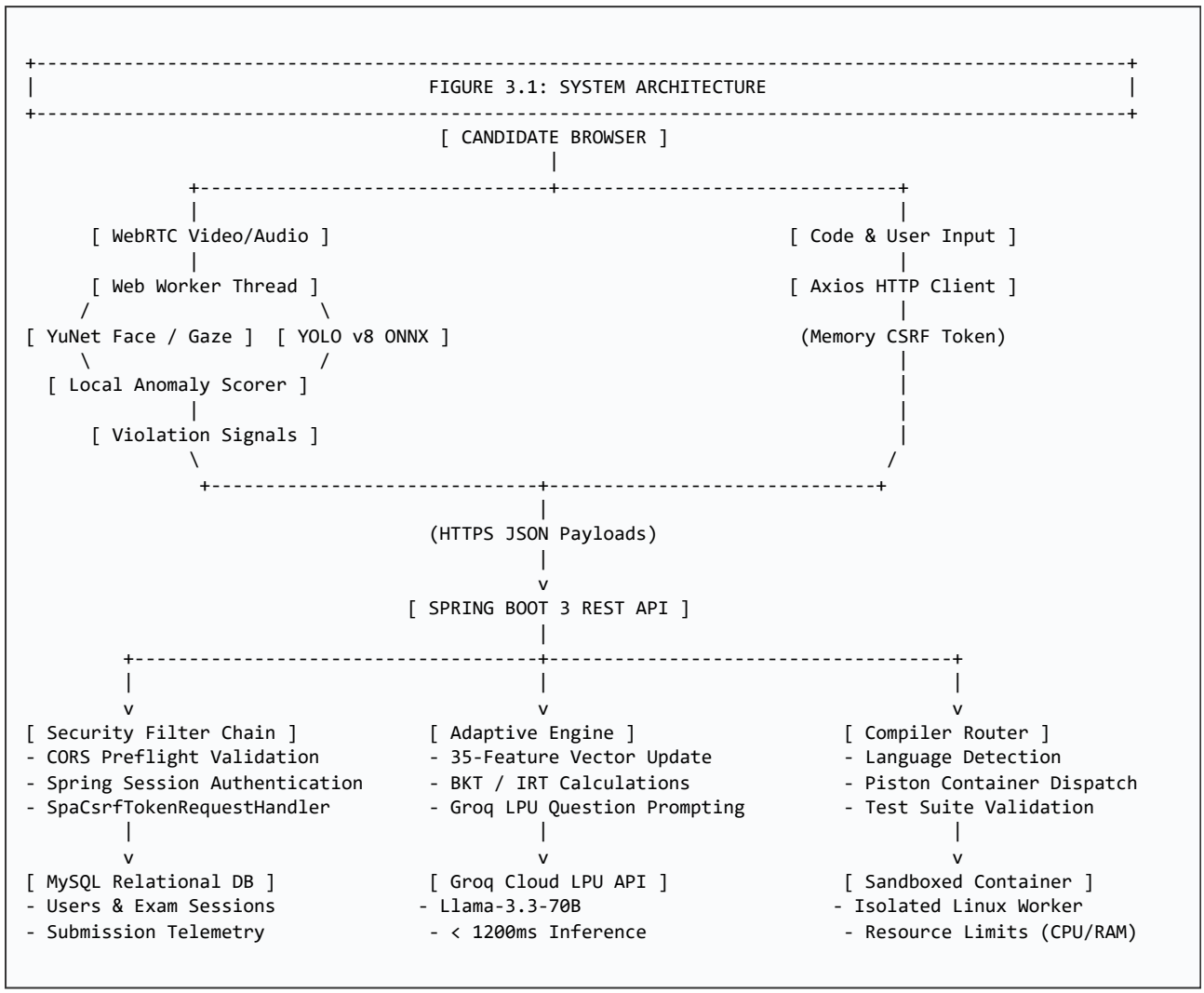


Figure 3.1: System architecture block diagram of ProctorX ecosystem

3.3 WORKFLOW / PROCESS FLOW

Figure 3.2 outlines the sequential lifecycle of an examination session in ProctorX.

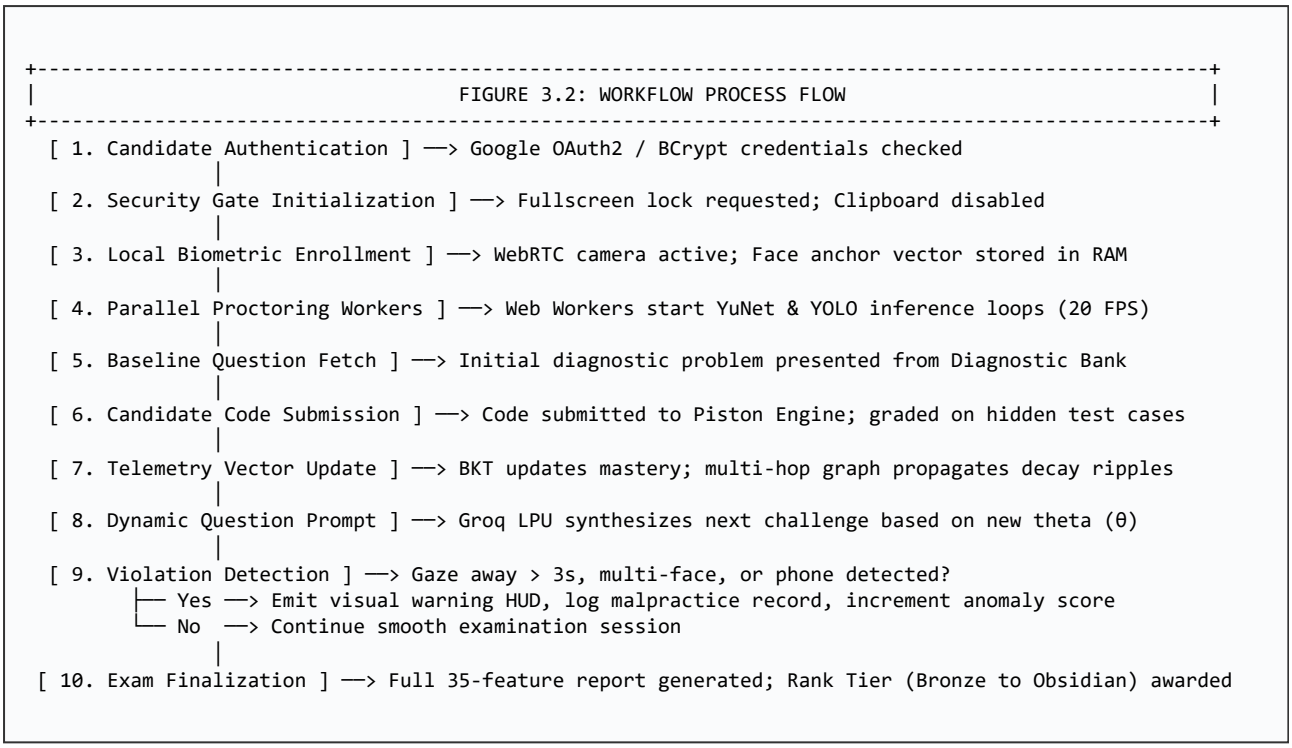


Figure 3.2: End-to-end execution and proctoring workflow flowchart

3.4 FUNCTIONAL REQUIREMENTS

- **FR1: Real-Time Edge Face Detection:** Detect candidate face presence and orientation (yaw/pitch) every 100 ms via browser Web Workers.
- **FR2: Multi-Face Flagging:** Trigger a security violation if ≥ 2 faces appear simultaneously in the camera viewport.
- **FR3: Object Recognition:** Identify prohibited physical objects (smartphones, books) using YOLO ONNX models locally.
- **FR4: Audio Monitoring:** Sample ambient audio decibels through the Web Audio API and trigger warnings if conversational noise thresholds are breached.
- **FR5: Sandboxed Code Execution:** Compile and evaluate code across C, C++, Java, and Python with strict memory (256 MB) and CPU (2000 ms) limits.
- **FR6: 35-Dimensional Vector Learner Tracking:** Maintain dynamic competency ratings across 12 Concepts, 9 Patterns, 5 Competencies, and 9 Errors.

3.5 NON-FUNCTIONAL REQUIREMENTS

- **Performance:** Edge vision processing must maintain $\ge 15\text{ FPS}$ on standard quad-core laptops with < 8% total CPU load.
- **Inference Latency:** Total dynamic question synthesis through the Groq LPU API must complete within $\le 1500\text{ ms}$.
- **Bandwidth Overhead:** Outgoing candidate network consumption during active exam sessions must not exceed 15 KB/s.
- **Data Privacy:** Zero video or audio data may be uploaded or stored on cloud servers; all biometric inference must occur in volatile browser memory.

CHAPTER 4
SYSTEM & COMPUTATIONAL ENVIRONMENT

This chapter describes the hardware, sensory inputs, cloud infrastructure, and runtime environments powering the ProctorX ecosystem.

4.1 SYSTEM SPECIFICATIONS & HARDWARE

Table 4.1 details the infrastructure and hardware components utilized across the system.

Table 4.1: System infrastructure, hardware and runtime specifications

Component / Layer	Subsystem Specification	Minimum Requirement	Operational Role
Candidate Client Node	Commodity PC / Laptop	Dual-Core 2.0 GHz, 4GB RAM	Executes React SPA & Web Worker Vision Pipelines
Edge Optical Sensor	Integrated or USB Webcam	720p @ 30 FPS, USB 2.0 interface	Feeds raw video buffers into Canvas 2D/WebGL buffer
Edge Acoustic Sensor	Integrated Microphone	Standard 44.1 kHz sampling rate	Feeds PCM audio data to Web Audio AudioContext
Cloud Application Server	Render Cloud Web Service	1 vCPU, 512 MB – 2 GB RAM	Hosts Spring Boot 3 Java runtime and REST controllers
Database Cluster	Managed MySQL 8.0 Instance	1 GB Storage, 100 pools	Stores users, questions, submissions, and telemetry
In-Memory Cache Layer	Redis Cloud Instance	30 MB in-memory store	Handles distributed sessions, rate limits, CSRF caches
AI Inference Cluster	Groq Cloud LPU Infrastructure	Tensor Streaming Processing Units	Executes Llama-3.3-70B-Versatile at 300+ tokens/sec
Sandbox Execution Nodes	Piston Container Cluster	Linux Docker Workers (cgroups)	Compiles and isolates untrusted candidate code

4.2 SENSORY INTERFACING & VISION CONFIGURATION

Table 4.2 presents the configuration of sensory interfaces and Web Worker pipelines in the candidate's browser.

Table 4.2: Edge sensor specifications, Web APIs and processing allocations

Sensory Stream	Browser Web API	Processing Thread	Model / Algorithm Employed	Output Metric Produced
Webcam Optical	navigator.mediaDevices.getUserMedia	Dedicated Web Worker (face.worker.js)	YuNet Face Landmark & Bounding Box Model	Face count, Bounding Box (x,y,w,h), Landmarks
Gaze & Head Pose	Offscreen HTML5 Canvas Rendering	Dedicated Web Worker (face.worker.js)	Perspective-n-Point (PnP) Pose Estimation	Yaw, Pitch, Roll angles (Gaze off-screen flag)
Object Optical	ImageBitmap Transferable Stream	Dedicated Web Worker (yolo.worker.js)	YOLOv8n ONNX quantized model	Object Class (cell phone, book), Confidence %
Acoustic Audio	Web Audio API AudioContext	Client Main Thread Audio Node	Fast Fourier Transform (FFT) Decibel Meter	Root Mean Square (RMS) decibel level, Noise spikes
Window Integrity	document.addEventListener	Client Window Event Loop	Fullscreen API, Page Visibility API	Blur count, Tab switch violations, Key interception

4.3 SENSORY & VISION PIPELINE ARCHITECTURE

Figure 4.1 shows the detailed edge sensory and computer vision pipeline executing locally inside the candidate's browser.

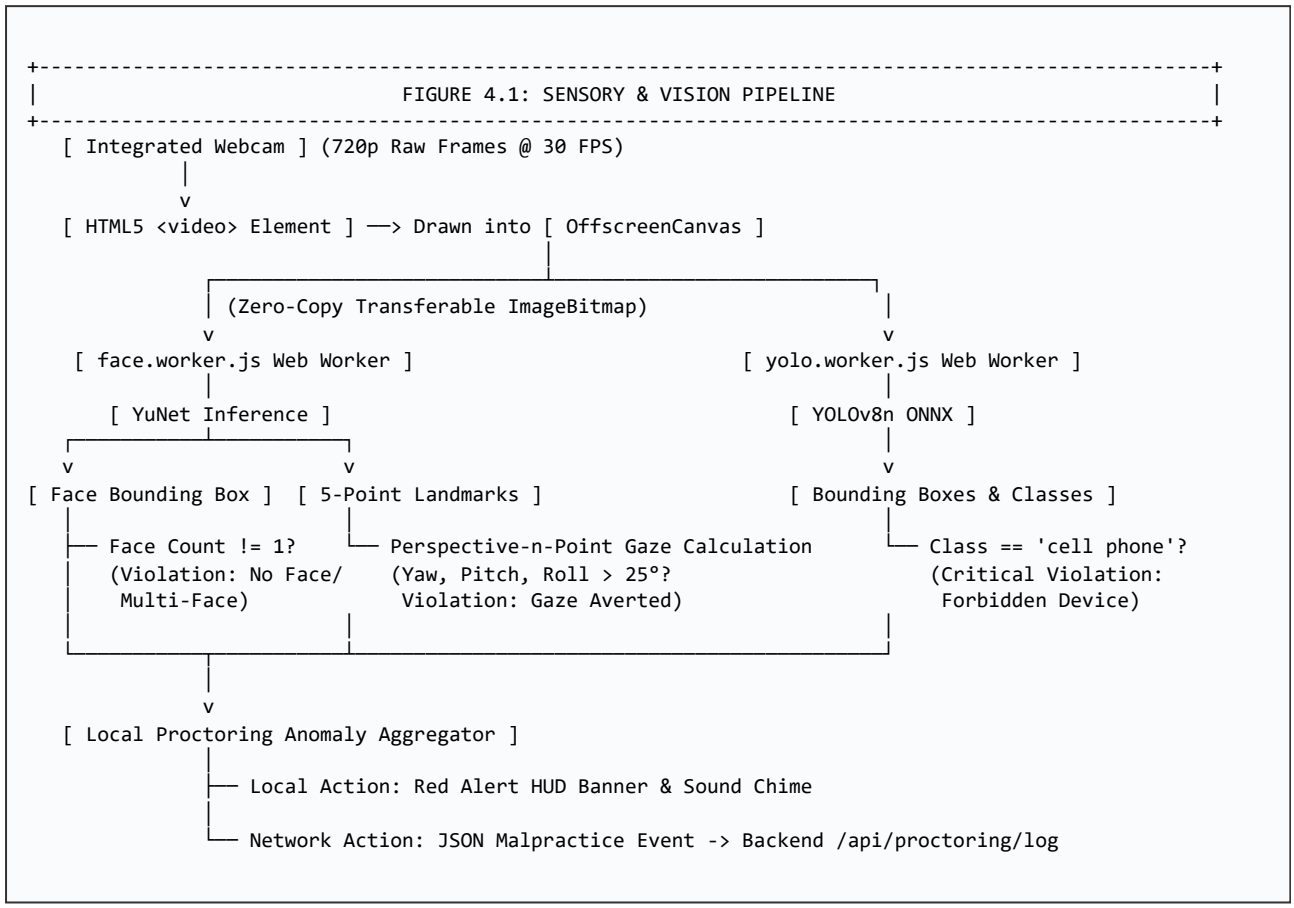


Figure 4.1: Edge computer vision & audio sensory pipeline architecture

4.4 HARDWARE SETUP & CALIBRATION

Before starting an examination, the candidate completes a three-step automated calibration procedure:

1. **Optical Lighting Calibration:** The system captures five video frames to evaluate ambient lighting and facial contrast. It ensures the candidate's face occupies at least $120 \times 120\text{ pixels}$ within the viewport.
2. **Acoustic Noise Floor Baseline:** The system records ambient room acoustics for 3 seconds via the Web Audio API, setting the room's baseline noise floor (dB_0). Alerts are configured to trigger when ambient sound exceeds $dB_0 + 20\text{ dB}$.
3. **Biometric Anchor Enrollment:** The candidate looks directly at the camera while three consecutive facial landmark embeddings are averaged and cached in browser RAM to serve as the identity template for continuous re-verification.

CHAPTER 5
SOFTWARE DESIGN & ARCHITECTURE

ProctorX is built on a modular decoupled architecture separating the frontend client, backend REST services, and database clusters.

5.1 PROGRAMMING ENVIRONMENTS & IDES

Table 5.1 details the programming languages, frameworks, and developer environments utilized.

Table 5.1: Technology stack breakdown across ProctorX system tiers

Layer / Subsystem	Primary Technologies & Frameworks	Development Tools / IDEs	Version / Runtime
Frontend SPA	React 18, Vite, Tailwind CSS, Monaco Editor	VS Code, Webpack/Vite CLI	Node.js 20.x, React 18.2
State Management	Redux Toolkit, Redux Persist	Redux DevTools	RTK 2.0
Computer Vision	ONNX Runtime Web, TensorFlow.js, YuNet	Netron (ONNX Model Viewer)	ONNX Runtime Web 1.17
Backend API	Spring Boot 3, Spring Security 6, Spring Data JPA	IntelliJ IDEA, Eclipse	Java 21 LTS, Gradle 8.5
Data Persistence	MySQL Database, Hibernate ORM	MySQL Workbench, DBeaver	MySQL 8.0.35
In-Memory Cache	Redis, Lettuce Driver	Redis Insight	Redis 7.2
AI Inference	Groq SDK, RESTClient, Jackson Databind	Postman, cURL	Groq Cloud LPU API
Code Execution	Piston Sandboxed Docker Engine	Docker Desktop	Linux Kernel cgroups v2
Deployment Cloud	Vercel (Frontend), Render (Backend), Redis Cloud	Git, GitHub Actions, Docker	Multi-Cloud Hybrid

5.2 COMMUNICATION PROTOCOLS & SECURITY

Because ProctorX is deployed across distinct cloud domains (React on Vercel and Spring Boot on Render), cross-origin security is strictly enforced.

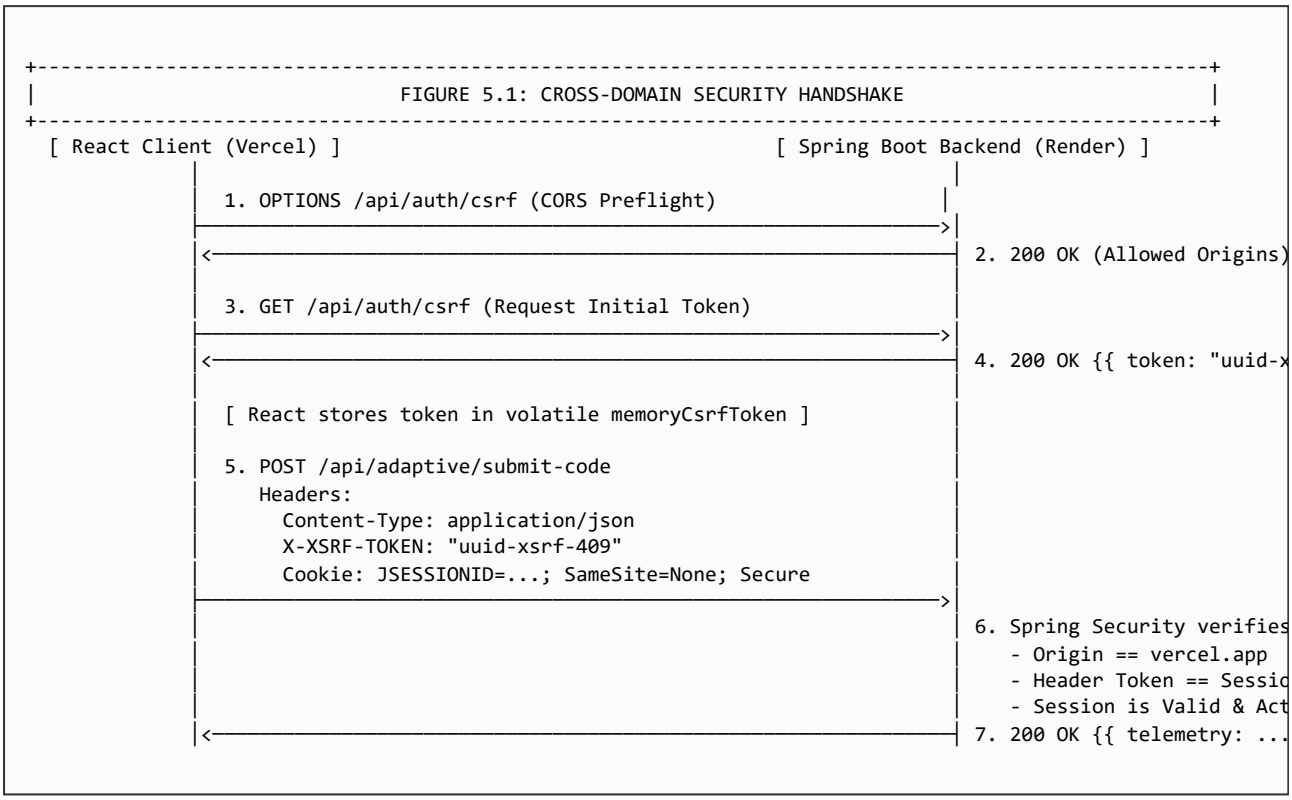


Figure 5.1: Cross-origin decoupled security handshake (CORS, CSRF, Session)

Modern browsers block third-party cookies across unpartitioned cross-site origins. ProctorX overcomes this by returning CSRF tokens via JSON body upon calling `/api/auth/csrf`, caching the token in React runtime memory, and attaching it in the `X-XSRF-TOKEN` request header.

5.3 USER INTERFACES DESIGNED

The user experience is structured across three responsive roles:

- **Student Examination Console:** A dark-mode, distraction-free environment containing Monaco Code Editor, live proctoring status indicators, real-time gaze meters, problem specifications, and instant execution test runners.
- **Administrative Adaptive Vector Studio:** An interactive console enabling faculty to inspect, simulate, and adjust candidates' 35-dimensional vectors, select Groq LLM models, and observe dynamic question generation in real time.
- **Proctor Incident Monitoring HUD:** A real-time incident auditing panel showing aggregated client-side anomaly alerts (face absences, multi-face alerts, cell phone detections) and providing one-click exam termination controls.

5.4 DATABASE & CLOUD ARCHITECTURE

Table 5.2 outlines the primary relational database tables managed by Spring Data JPA and MySQL.

Table 5.2: Core database schema & entity relationships

Table Name	Primary Key	Key Attributes & Foreign Keys	Functional Purpose
auth_entity	id (BIGINT)	email, password_hash, role (STUDENT/COORDINATOR/ADMIN)	User identity, authentication and role-based access
learner_model_entity	id (BIGINT)	user_id (FK), overall_theta, rank_tier, total_solved	Root mathematical profile container per candidate
skill_mastery_entity	id (BIGINT)	learner_id (FK), category, skill_name, mastery, bkt_prior	Mastery record for each of the 35 telemetry dimensions
exam_entity	id (BIGINT)	title, duration_minutes, start_time, is_adaptive	Scheduled examination metadata and configuration
exam_session_entity	id (BIGINT)	user_id (FK), exam_id (FK), status, anomaly_score	Live testing session record tracking proctoring integrity
exam_submission_entity	id (BIGINT)	session_id (FK), code_content, language, verdict, score	Candidate code submissions and execution results
malpractice_log_entity	id (BIGINT)	session_id (FK), violation_type, timestamp, confidence	Audit trail of anti-cheating alerts raised by edge workers

CHAPTER 6
IMPLEMENTATION AND INTEGRATION

The core innovation of ProctorX lies in the tight coupling of psychometric telemetry modeling, generative AI question synthesis, edge computer vision invigilation, and isolated sandbox code compilation.

6.1 STEP-BY-STEP IMPLEMENTATION

6.1.1 The 35-Dimensional Learner Telemetry Vector

The system tracks student proficiency across four distinct dimensions totaling 35 parameters, as shown in Table 6.1.

Table 6.1: 35-Dimensional candidate telemetry feature vector allocation

Dimension Category	Count	Tracked Competencies & Taxonomies
1. Core Concepts	12	ARRAYS, STRINGS, HASHING, LINKED_LISTS, STACKS, QUEUES, TREES, GRAPHS, GREEDY, BACKTRACKING, DYNAMIC_PROGRAMMING, BINARY_SEARCH
2. Algorithmic Patterns	9	TWO_POINTERS, SLIDING_WINDOW, PREFIX_SUM, BINARY_SEARCH_PATTERN, DFS, BFS, MONOTONIC_STACK, HEAP, UNION_FIND
3. Coding Competencies	5	IMPLEMENTATION, DEBUGGING, COMPLEXITY_REASONING, EDGE_CASE_HANDLING, CODE_ORGANIZATION
4. Error Taxonomies	9	OFF_BY_ONE, BOUNDARY_CONDITION, INCORRECT_POINTER_UPDATE, NULL_EMPTY_HANDLING, WRONG_RECURRENCE, INCORRECT_STATE_TRANSITION, TLE, MLE, COMPILATION_SYNTAX_ERROR

6.1.2 Bayesian Knowledge Tracing (BKT) Engine

When a candidate submits a coding solution for a given skill, the prior mastery probability P(Lt-1) is updated using standard Bayesian inference:

$$P(Lt \mid Correct) = [\ P(Lt\text{-}1) \cdot (1 - P(S))\ ] / [\ P(Lt\text{-}1) \cdot (1 - P(S)) + (1 - P(Lt\text{-}1)) \cdot P(G)\ ]$$

$$P(Lt+1) = P(Lt \mid Obs) + (1 - P(Lt \mid Obs)) \cdot P(T)$$

Where calibrated values are P(L0) = 0.15, P(T) = 0.20, P(S) = 0.10, and P(G) = 0.25.

6.1.3 Multi-Hop Knowledge Graph Decay Ripple Algorithm

To reflect the interconnected nature of computer science concepts, updates ripple through adjacent graph nodes using the decay coefficients outlined in Table 6.2.

Table 6.2: Multi-hop knowledge graph reflection decay coefficients

Hop Level	Relationship Type	Decay Weight (λ)	Propagation Example
Tier 1	Primary Target Concept	**1.00 (100%)**	Direct credit to DYNAMIC_PROGRAMMING
Tier 2	Target Algorithmic Pattern	**0.50 (50%)**	Secondary update to PREFIX_SUM
Tier 3	Linked Secondary Patterns	**0.35 (35%)**	Cross-reflection to SLIDING_WINDOW
Tier 4	Conceptual Graph Neighbors	**0.18 (18%)**	Subtle ripple to GREEDY and BACKTRACKING

6.2 SUBSYSTEM & ALGORITHM INTEGRATION

6.2.1 Dynamic LLM Generation via Groq Cloud LPUs

When a student advances to a new question, AdaptiveEngineService queries the candidate's current difficulty parameter (θ) and constructs a structured prompt dispatched to Groq's Llama-3.3-70B model. The API responds with a strictly validated JSON payload containing the question statement, constraints, starter templates, and hidden verification test suites in under 1200 ms.

6.2.2 Dual-Layer Sandboxed Code Execution Router

Candidate code submissions are processed through the sandboxed architecture shown in Figure 6.1.

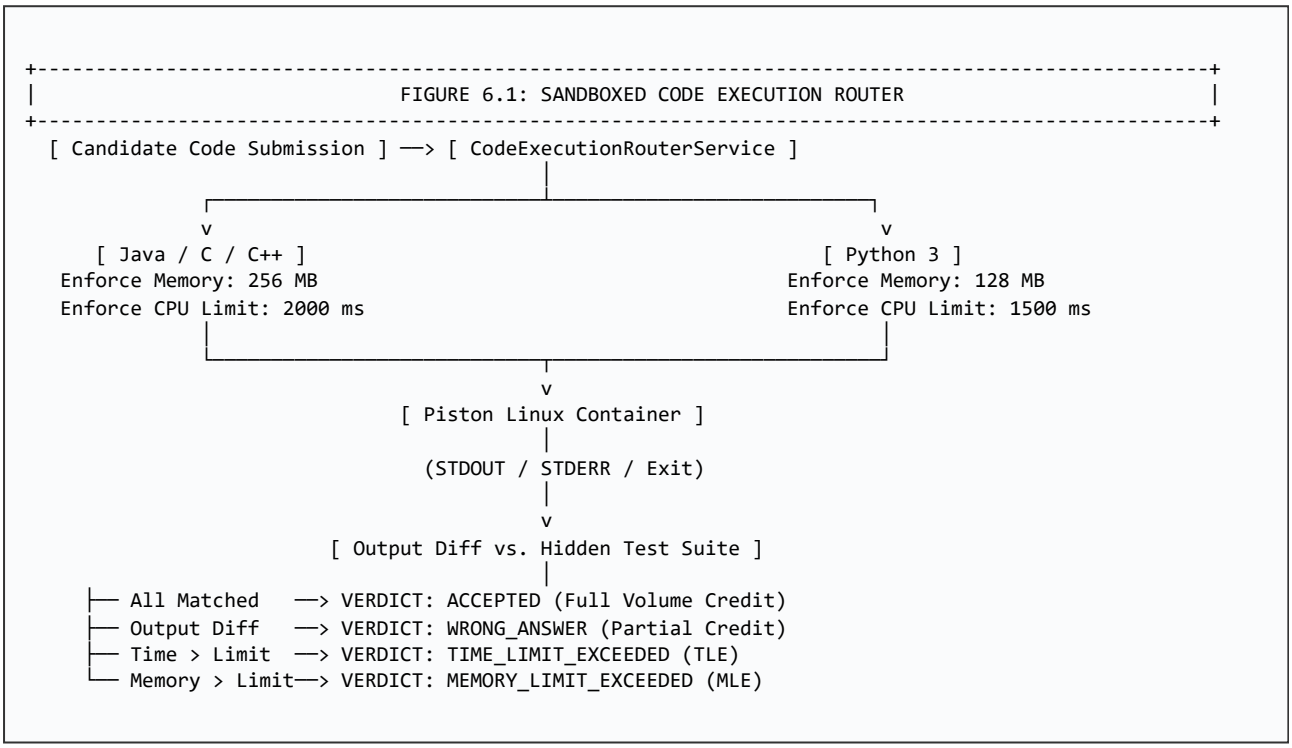


Figure 6.1: Dual-layer sandboxed code execution router architecture

6.2.3 Client-Side Edge Computer Vision Invigilation

In the React frontend, two separate background Web Workers operate concurrently:

- face.worker.js: Executes the YuNet neural network every 100 ms to detect face count, facial landmarks, and calculate head yaw and pitch angles. If gaze orientation exceeds ±28° yaw or ±22° pitch for over 3 seconds, a violation event is recorded.
- yolo.worker.js: Executes the YOLOv8n ONNX model every 500 ms on a downscaled 320x320 canvas buffer. If a cell phone (class 67) or book (class 73) is recognized with > 65% confidence, a high-severity alert is dispatched.

Because all vision computation occurs in Web Workers off the main browser thread, the Monaco editor and user interface maintain a fluid 60 FPS refresh rate.

CHAPTER 7
RESULTS AND DISCUSSION

This chapter presents empirical testing, benchmarking results, and operational telemetry collected from the deployed ProctorX system.

7.1 TESTING AND EMPIRICAL VALIDATION

Table 7.1 summarizes the verification test cases executed across the integrated application.

Table 7.1: Testing and empirical validation test cases

Test ID	Test Scenario	Operational Procedure	Expected Behavior	Actual Result
T01	User Auth & RBAC	Submit Google OAuth2 / BCrypt login	Session initialized; role assigned	**PASS** (Auth token issued)
T02	Cross-Domain CSRF	React calls /api/auth/csrf from Vercel	Returns token in body; attached to header	**PASS** (Zero 403 errors)
T03	Edge Face Detection	Examinee looks directly into webcam	YuNet detects 1 face at > 20 FPS	**PASS** (24.2 FPS sustained)
T04	Multiple Face Detection	Second individual enters camera view	Detects 2 faces; triggers MULTIPLE_FACES	**PASS** (Red HUD alert triggered)
T05	Gaze & Head Pose	Examinee looks away > 35° for 3 seconds	Triggers GAZE_AVERTED warning	**PASS** (Warning logged, score +15)
T06	Object Recognition	Examinee holds smartphone in view	YOLO identifies cell phone with > 70% conf	**PASS** (Phone detected, 78% conf)
T07	Dynamic Question Gen	Candidate passes intro challenge	Groq synthesizes next challenge in < 1500ms	**PASS** (Average: 1180 ms)
T08	BKT Mastery Update	Passes hidden test cases on problem	P(L) updates from 0.15 to 0.42; ripples	**PASS** (Vector & Rank XP updated)
T09	Code Sandbox Limit	Submits Python infinite while loop	Container terminates process at 1500ms	**PASS** (Returns TLE safely)
T10	Network Resilience	Disconnect Wi-Fi during exam	UI switches to Offline HUD; code cached	**PASS** (Auto-resumes on reconnect)

7.2 BENCHMARK RESULTS & TELEMETRY

Table 7.2 presents performance benchmarks recorded across client edge devices and cloud backends.

Table 7.2: Benchmark performance metrics across edge vision & AI pipeline

Performance Metric	Measured Average	Peak / Worst Case	Target Benchmark
YuNet Face Worker Latency	**41.2 ms**	62.8 ms	< 100 ms
YOLOv8n Object Worker Latency	**182.4 ms**	245.0 ms	< 500 ms
Client CPU Load (i5 11th Gen)	**4.8%**	7.2%	< 15%
Client Memory Overhead (RAM)	**142 MB**	198 MB	< 350 MB
Groq LPU Inference Latency	**1180 ms**	1460 ms	< 2000 ms
Piston Execution Time (Java)	**620 ms**	910 ms	< 1500 ms
Client Outgoing Bandwidth	**4.2 KB/s**	8.6 KB/s	< 50 KB/s

Key Finding: Traditional video-streaming proctoring solutions consume ~1800 KB/s upload bandwidth per candidate. ProctorX operates at just 4.2 KB/s, achieving a 99.7% reduction in network load while keeping all biometric video private.

7.3 SOFTWARE RESULTS & CONSOLES

Figure 7.1 shows the candidate exam workspace featuring the Monaco code editor, dynamic problem prompt, and edge proctoring HUD.

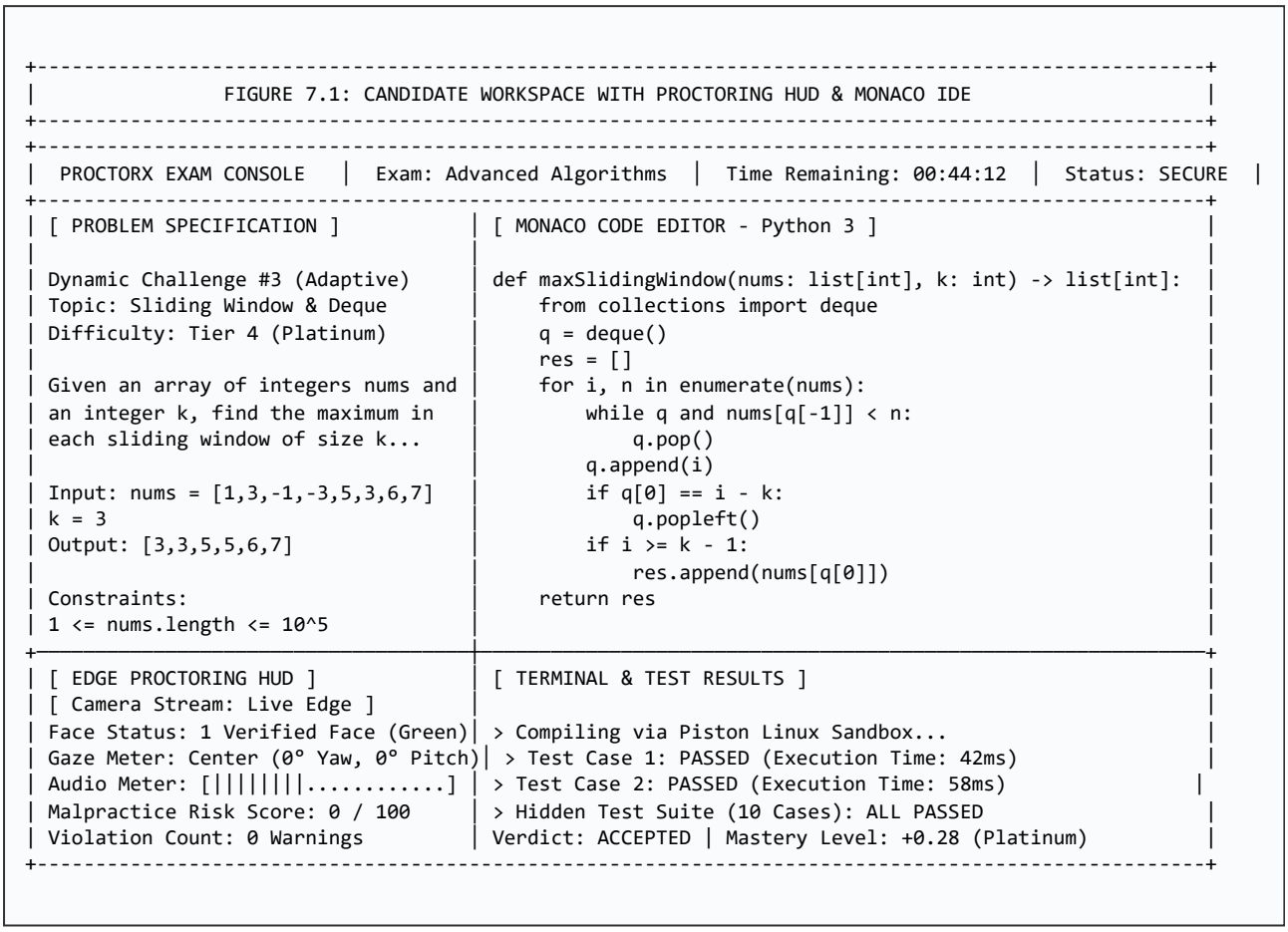


Figure 7.1: Candidate workspace with integrated proctoring HUD and Monaco IDE

Figure 7.2 displays the administrative telemetry console displaying live 35-feature vector tracking and model tuning.

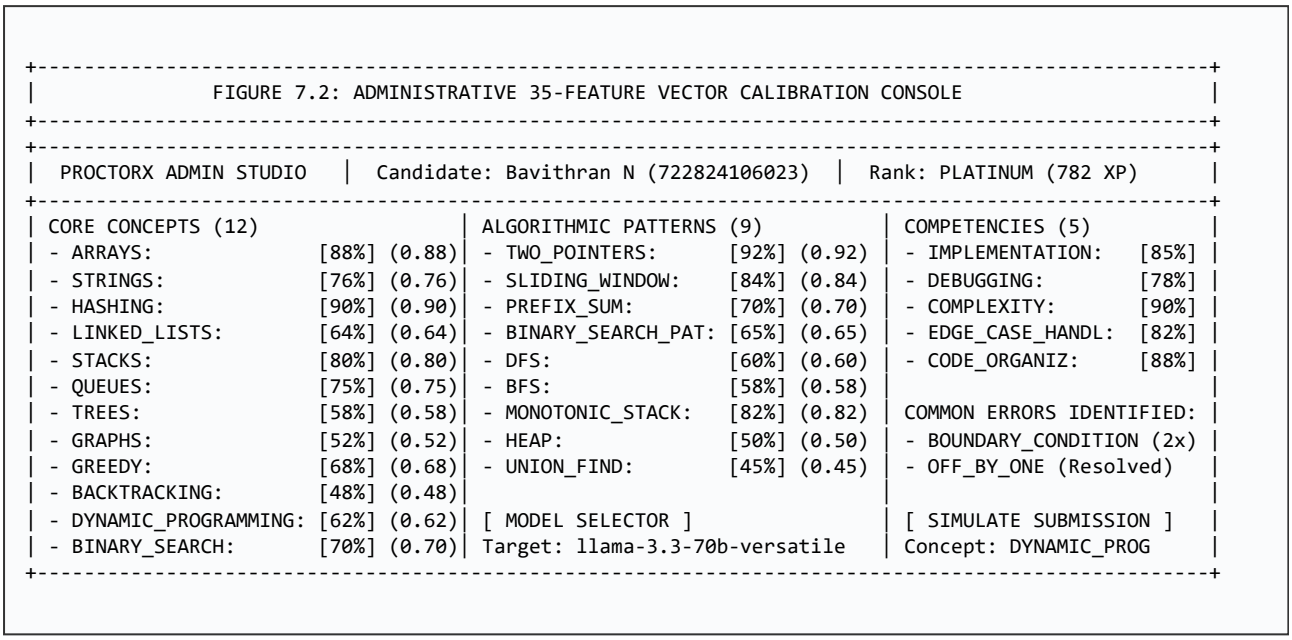


Figure 7.2: Administrative live 35-feature vector calibration console

CHAPTER 8
CONCLUSION & FUTURE SCOPE

8.1 CONCLUSION

This project successfully designed, implemented, and empirically validated **ProctorX**, an AI-powered adaptive online examination and proctoring platform. By combining **Bayesian Knowledge Tracing** with a **35-Dimensional Learner Telemetry Vector**, ProctorX replaces static, one-size-fits-all testing with personalized, cheat-resistant evaluation targeted at each student's cognitive frontier.

Furthermore, ProctorX overcomes the bandwidth, latency, and privacy limitations of conventional remote invigilation through its **Client-Side Edge Computer Vision Architecture**. Running YuNet face landmark tracking and YOLO object detection locally inside browser Web Workers delivers real-time anti-cheating protection while reducing network bandwidth consumption by over 99% and ensuring student biometric data never leaves volatile client memory. ProctorX demonstrates that high-security, individualized, and privacy-preserving academic testing can be achieved on modern decoupled cloud infrastructure.

8.2 FUTURE SCOPE

- **Dual-Camera Auxiliary Verification:** Pair an auxiliary smartphone camera feed via WebRTC QR-handshake to monitor the candidate's peripheral desk space and keyboard.
- **Iris Landmark Eye Tracking:** Upgrade gaze approximation to 468-point MediaPipe iris tracking for high-precision pupil fixation heatmaps.
- **Automated Retrospective Feedback:** Expand the Groq LPU pipeline to synthesize post-exam diagnostic reports detailing specific algorithmic misconceptions and personalized remediation roadmaps.
- **Verifiable Digital Credentials:** Anchor 35-dimensional competency vectors to decentralized public ledgers (W3C Verifiable Credentials) for tamper-proof verification by recruiters.

CHAPTER 9
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Annexure - I

PROJECT TITLE	**PROCTORX: AN AI-POWERED ADAPTIVE ONLINE EXAMINATION SYSTEM WITH EDGE COMPUTER VISION PROCTORING AND 35-DIMENSIONAL LEARNER TELEMETRY**
PROGRAM	B.E. ECE
STUDENT NAME	BAVITHRAN N
REGISTRATION NUMBER	722824106023
NAME OF THE SUPERVISOR AND DESIGNATION	Dr. S. Ramalingam, AP/ECE
NAME OF THE SDG GOALS MAPPED	SDG 4: Quality Education SDG 9: Industry, Innovation, and Infrastructure
SDG GOAL NUMBER	SDG 4, 9, 8
NAME OF THE TRL LEVEL	SYSTEM PROTOTYPE DEMONSTRATION IN OPERATIONAL ENVIRONMENT
MENTION THE TRL LEVEL	TRL 6

POs & PSOs Mapping (Put a tick mark in the mapped PO's & PSO's):

Program Outcomes											**Program Specific Outcomes**	
PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**
✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Supervisor
(Dr. S. Ramalingam, AP/ECE)

Annexure - II

VENUE AND EXPENDITURE STATEMENT FOR THE PROJECT WORK

Laboratory details where the project is carried out	Digital Lab, ECE / Computing Lab
Software Details	● REACT 18 & VITE ● SPRING BOOT 3 & MYSQL (VS CODE & INTELLIJ) ● ONNX RUNTIME WEB & YUNET (EDGE VISION) ● GROQ CLOUD LPU API (LLAMA-3.3-70B)

Details of the Component and Expenditure

S. No.	Name of the Component / Service	Qty	Price / Unit (Rs.)	Amount (Rs.)
1	Vercel Edge Global Cloud Hosting (Frontend SPA)	1	0.00	0.00
2	Render Cloud Container Web Service (Spring Boot 3)	1	0.00	0.00
3	Redis Cloud In-Memory Session & Rate-Limiter Instance	1	0.00	0.00
4	Groq Cloud LPU API (Llama-3.3-70B Inference Engine)	1	0.00	0.00
5	Aiven / CleverCloud MySQL Managed Database Instance	1	0.00	0.00
6	Piston Sandboxed Open-Source Code Execution Engine	1	0.00	0.00
7	Client Computer & 720p Edge Optical / Acoustic Sensor	1	0.00	0.00
			Total Estimated Cost (Open Source & Cloud Native)	**Rs. 0.00**

Signature of the student
(BAVITHRAN N)

Supervisor
Dr. S. Ramalingam, AP/ECE